

AI ASSISTED CODING

LAB TEST-3

P. LEELA PRASEEDA SAI

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BATCH-03

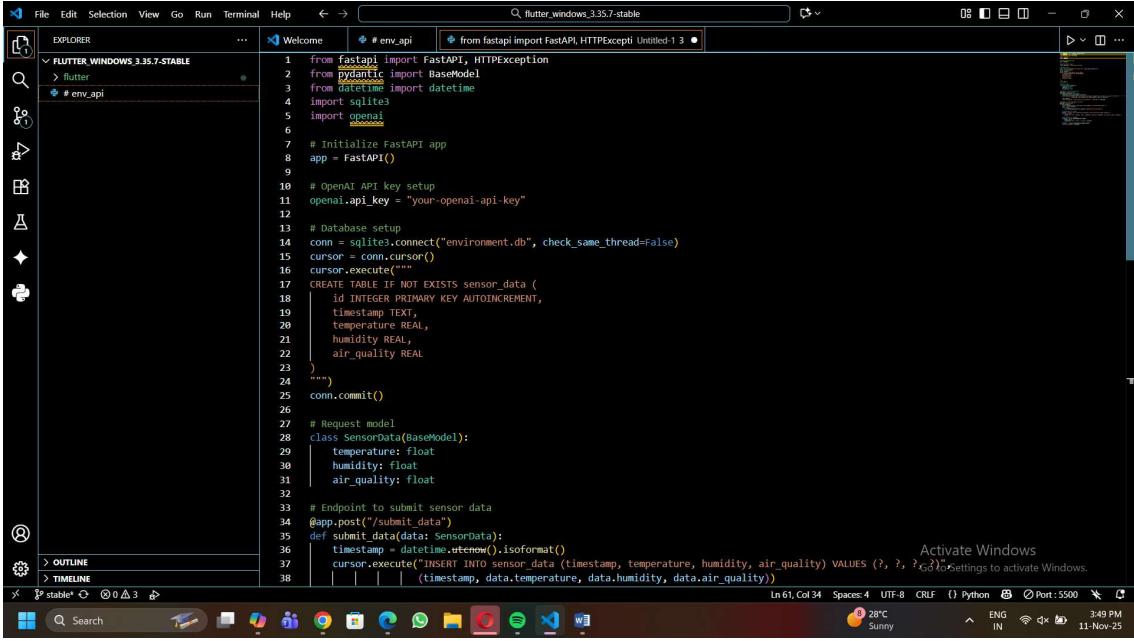
SET-E13

Q1.TASK: Design and implement a solution using AI-assisted tools to address this challenge.

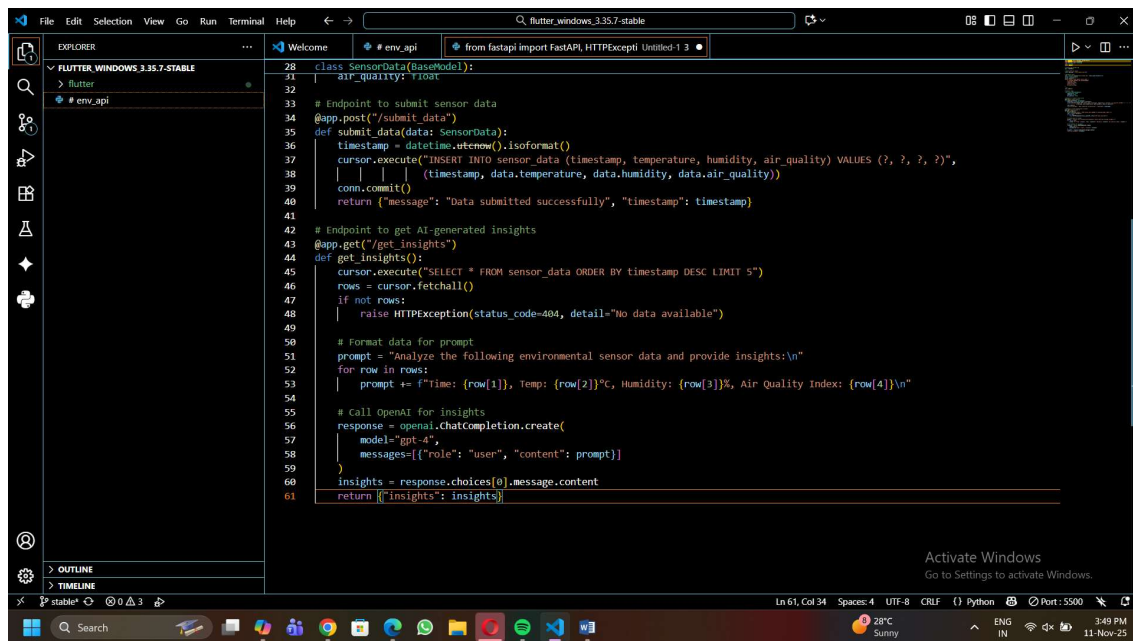
PROMPT:

Design and implement a backend API system that uses **AI-assisted tools** to detect anomalies in sensor data. automatically detect abnormal readings and provide simple explanations for them.

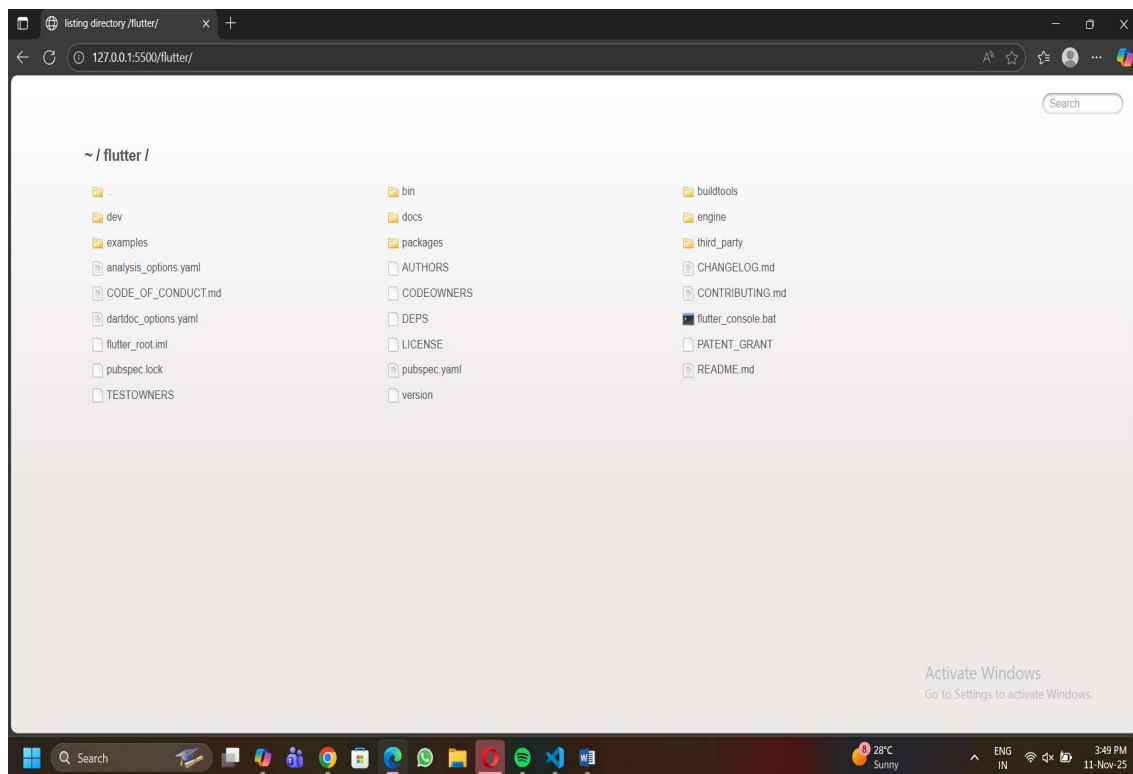
CODE:



```
1 from fastapi import FastAPI, HTTPException
2 from pydantic import BaseModel
3 from datetime import datetime
4 import sqlite3
5 import openai
6
7 # Initialize FastAPI app
8 app = FastAPI()
9
10 # OpenAI API key setup
11 openai.api_key = "your-openai-api-key"
12
13 # Database setup
14 conn = sqlite3.connect("environment.db", check_same_thread=False)
15 cursor = conn.cursor()
16 cursor.execute("""
17 CREATE TABLE IF NOT EXISTS sensor_data (
18     id INTEGER PRIMARY KEY AUTOINCREMENT,
19     timestamp TEXT,
20     temperature REAL,
21     humidity REAL,
22     air_quality REAL
23 )
24 """)
25 conn.commit()
26
27 # Request model
28 class SensorData(BaseModel):
29     temperature: float
30     humidity: float
31     air_quality: float
32
33 # Endpoint to submit sensor data
34 @app.post("/submit_data")
35 def submit_data(data: SensorData):
36     timestamp = datetime.utcnow().isoformat()
37     cursor.execute("INSERT INTO sensor_data (timestamp, temperature, humidity, air_quality) VALUES (?, ?, ?, ?)",
38                   (timestamp, data.temperature, data.humidity, data.air_quality))
```



OUTPUT:



EXPLANATION:

This system helps a company **collect and analyze environmental data** like temperature, humidity, and air quality. It uses:

- **FastAPI** to build the backend (a fast Python web framework)
- **SQLite** to store the data
- **OpenAI** to generate smart insights from recent sensor readings.

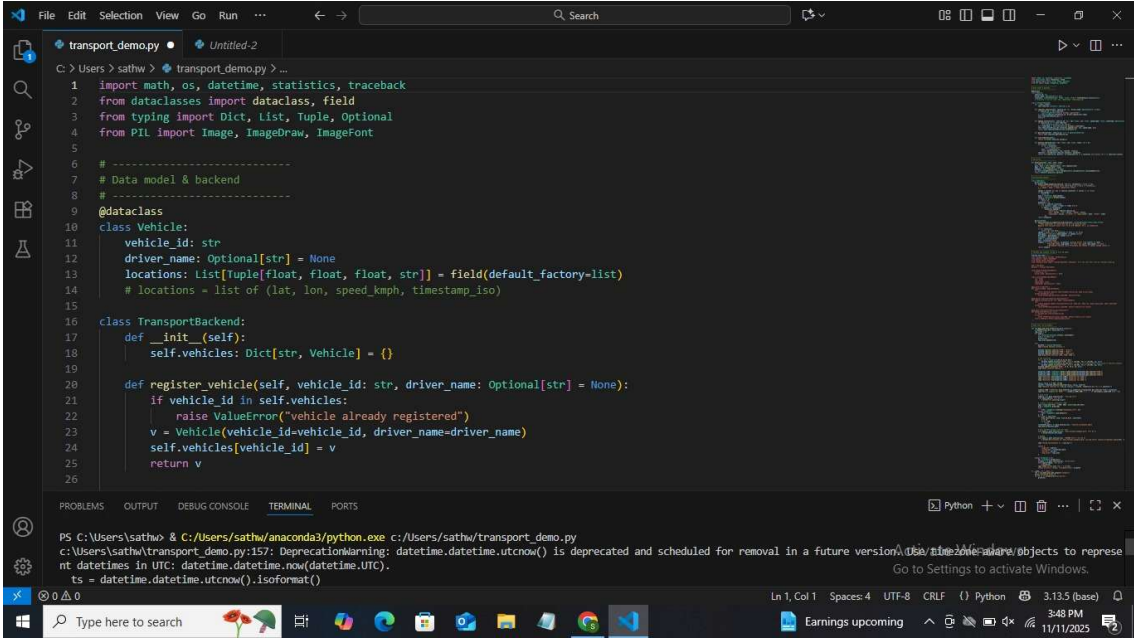
Q2. TASK:

Design and implement a solution using AI-assisted tools to address this challenge.

Include code, explanation of AI integration, and test results.

PROMPT: write a python code thar must given a backend solution for a transportation-related problem i.e In the domain of Transportation, a company is facing a challenge related to backend api development.

CODE:



```
transport_demo.py • Untitled-2
C:\Users\sathw> cd transport_demo.py > ...
1 import math, os, datetime, statistics, traceback
2 from dataclasses import dataclass, field
3 from typing import Dict, List, Tuple, Optional
4 from PIL import Image, ImageDraw, ImageFont
5
6 # -----
7 # Data model & backend
8 # -----
9 @dataclass
10 class Vehicle:
11     vehicle_id: str
12     driver_name: Optional[str] = None
13     locations: List[Tuple[float, float, float, str]] = field(default_factory=list)
14     # locations = list of (lat, lon, speed_kmph, timestamp_iso)
15
16 class TransportBackend:
17     def __init__(self):
18         self.vehicles: Dict[str, Vehicle] = {}
19
20     def register_vehicle(self, vehicle_id: str, driver_name: Optional[str] = None):
21         if vehicle_id in self.vehicles:
22             raise ValueError("vehicle already registered")
23         v = Vehicle(vehicle_id=vehicle_id, driver_name=driver_name)
24         self.vehicles[vehicle_id] = v
25         return v
26
27
28 PS C:\Users\sathw> & C:\Users\sathw\anaconda3\python.exe c:\Users\sathw\transport_demo.py
c:\Users\sathw\transport_demo.py:157: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC).
  ts = datetime.datetime.utcnow().isoformat()
Go to Settings to activate Windows.
```

```
File Edit Selection View Go Run ... Search
transport_demo.py • Untitled-2
C:\Users\sathw > transport_demo.py ...
16 class TransportBackend:
25     return v
26
27 def update_location(self, vehicle_id: str, lat: float, lon: float, speed_kmph: float, timestamp: Optional[str] = None):
28     if vehicle_id not in self.vehicles:
29         raise KeyError("vehicle not found")
30     ts = timestamp or datetime.datetime.utcnow().isoformat()
31     self.vehicles[vehicle_id].locations.append((lat, lon, speed_kmph, ts))
32     return self.vehicles[vehicle_id].locations[-1]
33
34 def get_vehicle(self, vehicle_id: str) -> Optional[Vehicle]:
35     return self.vehicles.get(vehicle_id)
36
37 def list_vehicles(self):
38     return list(self.vehicles.values())
39
40 def nearest_vehicles(self, lat: float, lon: float, limit: int = 3):
41     def dist(v: Vehicle):
42         if not v.locations:
43             return float("inf")
44         last = v.locations[-1]
45         return haversine(lat, lon, last[0], last[1])
46     vehicles = sorted(self.vehicles.values(), key=dist)
47     return [(v.vehicle_id, dist(v), (v.locations[-1] if v.locations else None)) for v in vehicles[:limit]]
48
49 # -----
50
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python + - - - - -
PS C:\Users\sathw> & C:\Users\sathw\anaconda3\python.exe c:\Users\sathw\transport_demo.py
c:\Users\sathw\transport_demo.py:157: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use time zone aware objects to represent
nt datetimes in UTC: datetime.datetime.now(datetime.UTC).
  ts = datetime.datetime.utcnow().isoformat()
Go to Settings to activate Windows.
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF () Python 3.13.5 (base)
3:49 PM 11/11/2025
```

```
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transport_demo.py • Untitled-2
C:\Users\sathw > transport_demo.py ...
52 def haversine(lat1, lon1, lat2, lon2):
55     dphi = math.radians(lat2 - lat1)
56     dlamba = math.radians(lon2 - lon1)
57     a = math.sin(dphi/2)**2 + math.cos(phi1)*math.cos(phi2)*math.sin(dlamba/2)**2
58     return 2*R*math.asin(math.sqrt(a))
59
60 # -----
61 # AI-assisted helpers
62 # -----
63 class AIHelpers:
64     @staticmethod
65     def detect_speed_anomalies(vehicle: Vehicle, threshold_z: float = 2.5):
66         """Return list of anomaly dicts where absolute z-score >= threshold_z.
67         z = (speed - mean) / stdev (population stdev).
68         """
69         speeds = [loc[2] for loc in vehicle.locations if loc[2] is not None]
70         if len(speeds) < 2:
71             return []
72         mean = statistics.mean(speeds)
73         stdev = statistics.pstdev(speeds)
74         if stdev == 0:
75             return []
76         anomalies = []
77         for loc in vehicle.locations:
78             z = (loc[2] - mean) / stdev if stdev else 0
79             if abs(z) >= threshold_z:

```

The image shows a Windows 10 desktop environment. The primary focus is the Visual Studio Code (VS Code) editor, which is open to a Python file named `transport_demo.py`. The code defines a class `AIHelpers` with two methods:

- `detect_speed_anomalies`: A method that takes a `vehicle` object and a `threshold_z` (default 2.5). It calculates the absolute value of `z` and if it exceeds the threshold, it appends a dictionary to the `anomalies` list. The dictionary contains `vehicle_id`, `lat`, `lon`, `speed`, `timestamp`, `z_score`, `mean`, and `stddev`.
- `mock_external_ai_summarize_trip`: A static method that takes a list of location tuples. It calculates average and maximum speeds, the number of points, and a summary string. The summary string is a formatted string containing the number of points, location points (start to end), average speed, and maximum speed.

The terminal window at the bottom shows a deprecation warning for `datetime.datetime.utcnow()`, which is scheduled for removal in a future version of Python. The warning suggests using `datetime.datetime.now(datetime.UTC)` instead.

The taskbar at the bottom of the screen displays various application icons, including the Start menu, search, and several open applications. The system clock in the bottom right corner shows the time as 3:49 PM on 11/11/2025.

The image shows a Windows 10 desktop with a Visual Studio Code (VS Code) editor open. The editor is displaying a Python file named `transport_demo.py`. The code is a FastAPI application that uses Pydantic models for data validation. It defines two endpoints: a POST endpoint for registering a vehicle and a GET endpoint for updating a vehicle's location. The code uses the `datetime` module to handle timestamps.

```

111 from typing import Optional
112 from transport_demo import TransportBackend, AIHelpers  # If you save this file as transport_demo.py
113
114 app = FastAPI()
115 backend = TransportBackend()
116
117 class RegisterModel(BaseModel):
118     vehicle_id: str
119     driver_name: Optional[str] = None
120
121 class LocationModel(BaseModel):
122     lat: float
123     lon: float
124     speed_kmph: float
125     timestamp: Optional[str] = None
126
127 @app.post("/register")
128 def register(body: RegisterModel):
129     try:
130         return backend.register_vehicle(body.vehicle_id, body.driver_name)
131     except ValueError as e:
132         raise HTTPException(status_code=400, detail=str(e))
133
134 @app.post("/vehicles/{vehicle_id}/location")
135 def update_loc(vehicle_id: str, body: LocationModel):
136     try:

```

The terminal at the bottom shows the command to run the application using `uvicorn`:

```

PS C:\Users\sathw> & C:\Users\sathw\anaconda3\python.exe c:\Users\sathw\transport_demo.py
c:\Users\sathw\transport_demo.py:157: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent
datetimes in UTC: datetime.datetime.now(datetime.UTC).
  ts = datetime.datetime.utcnow().isoformat()

```

The status bar at the bottom of the VS Code window shows the file encoding as UTF-8 and the Python interpreter set to the base environment.


```
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transport_demo.py • Untitled-2
C:\Users\sathw > transport_demo.py > ...

128 def register(body: RegisterModel):
129     try:
130         return backend.register_vehicle(body.vehicle_id, body.driver_name)
131     except ValueError as e:
132         raise HTTPException(status_code=400, detail=str(e))
133
134 @app.post("/vehicles/{vehicle_id}/location")
135 def update_loc(vehicle_id: str, body: LocationModel):
136     try:
137         return backend.update_location(vehicle_id, body.lat, body.lon, body.speed_kmph, body.timestamp)
138     except KeyError:
139         raise HTTPException(status_code=404, detail="vehicle not found")
140
141 @app.get("/vehicles/{vehicle_id}/anomalies")
142 def anomalies(vehicle_id: str):
143     v = backend.get_vehicle(vehicle_id)
144     if not v:
145         raise HTTPException(status_code=404, detail="vehicle not found")
146     return AIHelpers.detect_speed_anomalies(v)
147
148
149 # -----
150 # Demo test run & outputs
151 # -----
152 def run_demo_and_save_outputs(out_dir="outputs"):
153     os.makedirs(out_dir, exist_ok=True)
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```
File Edit Selection View Go Run ... Search
transport_demo.py • Untitled-2
C:\Users\sathw > transport_demo.py > ...

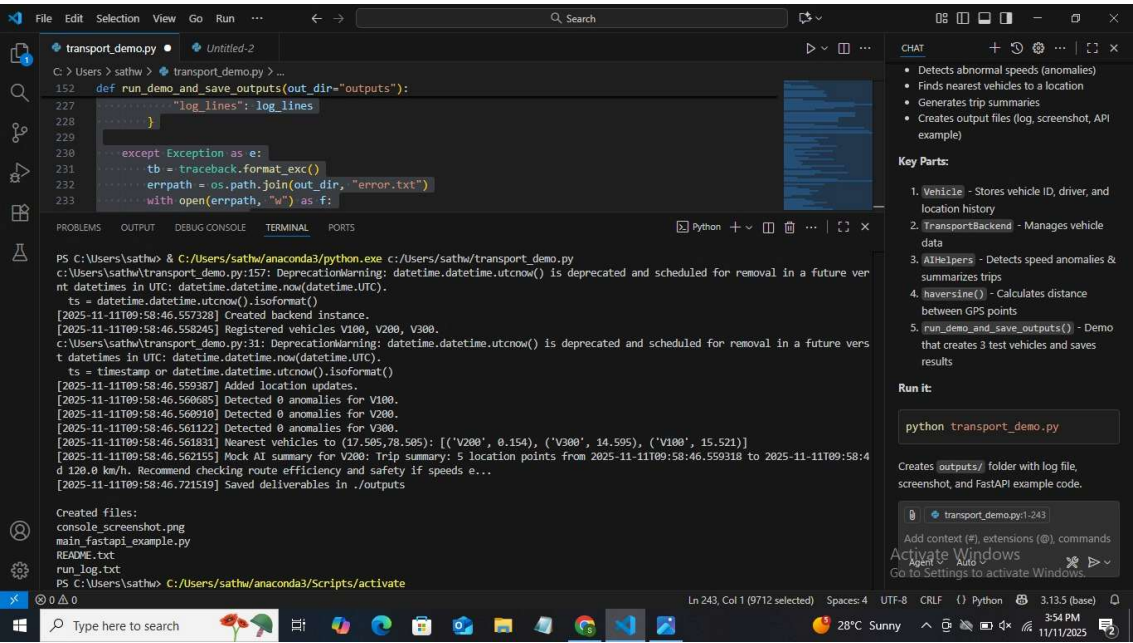
150 # Demo test run & outputs
151 # -----
152 def run_demo_and_save_outputs(out_dir="outputs"):
153     os.makedirs(out_dir, exist_ok=True)
154     log_lines = []
155     def log(s):
156         ts = datetime.datetime.utcnow().isoformat()
157         line = f"[{ts}] {s}"
158         print(line)
159         log_lines.append(line)
160
161     try:
162         backend = TransportBackend()
163         log("Created backend instance.")
164
165         backend.register_vehicle("V100", "Alice")
166         backend.register_vehicle("V200", "Bob")
167         backend.register_vehicle("V300", "Charlie")
168         log("Registered vehicles V100, V200, V300.")
169
170         # Add locations
171         for i, sp in enumerate([40,45,42,44,43]):
172             backend.update_location("V100", 17.4 + i*0.001, 78.4 + i*0.001, sp, None)
173         for i, sp in enumerate([30,32,31,120,33]): # 120 is an intentionally large value to simulate anomaly
174             backend.update_location("V200", 17.5 + i*0.001, 78.5 + i*0.001, sp, None)
175         backend.update_location("V300", 17.6, 78.6, 25, None)
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File Edit Selection View Go Run ... Search
transport_demo.py • Untitled-2
C:\Users\sathw> transport_demo.py ...
152 def run_demo_and_save_outputs(out_dir="outputs"):
153
154     # Add locations
155     for i, sp in enumerate([40,45,42,44,43]):
156         backend.update_location("V100", 17.4 + i*0.001, 78.4 + i*0.001, sp, None)
157     for i, sp in enumerate([30,32,31,120,33]): # 120 is an intentionally large value to simulate anomaly
158         backend.update_location("V200", 17.5 + i*0.001, 78.5 + i*0.001, sp, None)
159     backend.update_location("V300", 17.6, 78.6, 25, None)
160     log("Added location updates.")
161
162     anomalies_v100 = AIHelpers.detect_speed_anomalies(backend.get_vehicle("V100"))
163     anomalies_v200 = AIHelpers.detect_speed_anomalies(backend.get_vehicle("V200"))
164     anomalies_v300 = AIHelpers.detect_speed_anomalies(backend.get_vehicle("V300"))
165     log(f"Detected {len(anomalies_v100)} anomalies for V100.")
166     log(f"Detected {len(anomalies_v200)} anomalies for V200.")
167     log(f"Detected {len(anomalies_v300)} anomalies for V300.")
168
169     lat_q, lon_q = 17.505, 78.505
170     nearest = backend.nearest_vehicles(lat_q, lon_q, limit=3)
171     log(f"Nearest vehicles to ({lat_q},{lon_q}): {[n[0], round(n[1],3)} for n in nearest]")
172
173     summary_v200 = AIHelpers.mock_external_ai_summarize_trip(backend.get_vehicle("V200").locations)
174     log(f"Hack AI summary for V200: " + (summary_v200[:200] + ("..." if len(summary_v200)>200 else "")))
175
176     # Save files
177     logfile = os.path.join(out_dir, "run_log.txt")
178
179     PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
180 PS C:\Users\sathw> & C:\Users\sathw\anaconda3\python.exe c:\Users\sathw\transport_demo.py
181 c:\Users\sathw\transport_demo.py:157: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use time zone aware objects to represe
182 nt datetimes in UTC: datetime.datetime.now(datetime.UTC).
183 ts = datetime.datetime.utcnow().isoformat()
184
185 Ln 1, Col 1 Spaces: 4 UTF-8 CRLF () Python 3.13.5 (base)
186 Type here to search Earnings upcoming 3:49 PM 11/11/2025
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File Edit Selection View Go Run ... Search
transport_demo.py • Untitled-2
C:\Users\sathw> transport_demo.py ...
152 def run_demo_and_save_outputs(out_dir="outputs"):
153
154     log(f"Hack AI summary for V200: " + (summary_v200[:200] + ("..." if len(summary_v200)>200 else "")))
155
156     # Save files
157     logfile = os.path.join(out_dir, "run_log.txt")
158     with open(logfile, "w") as f:
159         f.write("\n".join(log_lines))
160
161     # create a console screenshot (PNG)
162     img = Image.new("RGB", (1200, 800), color=(255,255,255))
163     draw = ImageDraw.Draw(img)
164     try:
165         font = ImageFont.truetype("DejaVuSans.ttf", 14)
166     except Exception:
167         font = ImageFont.load_default()
168     y = 10
169     for line in log_lines:
170         draw.text((10,y), line, fill=(0,0,0), font=font)
171         y += 18
172         if y > 780:
173             break
174     screenshot_path = os.path.join(out_dir, "console_screenshot.png")
175     img.save(screenshot_path)
176
177     # Save FastAPI example file for user
178     with open(os.path.join(out_dir, "main_fastapi_example.py"), "w") as f:
179
180     PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
181 PS C:\Users\sathw> & C:\Users\sathw\anaconda3\python.exe c:\Users\sathw\transport_demo.py
182 c:\Users\sathw\transport_demo.py:157: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use time zone aware objects to represe
183 nt datetimes in UTC: datetime.datetime.now(datetime.UTC).
184 ts = datetime.datetime.utcnow().isoformat()
185
186 Ln 1, Col 1 Spaces: 4 UTF-8 CRLF () Python 3.13.5 (base)
187 Type here to search Earnings upcoming 3:50 PM 11/11/2025
```

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File Edit Selection View Go Run ... Search
transport_demo.py • Untitled-2
C:\Users\sathw> transport_demo.py ...
152 def run_demo_and_save_outputs(out_dir="outputs"):
153     img.save(screenshot_path)
154
155     # Save FastAPI example file for user
156     with open(os.path.join(out_dir, "main_fastapi_example.py"), "w") as f:
157         f.write(FASTAPI_APP_CODE)
158
159     # README
160     with open(os.path.join(out_dir, "README.txt"), "w") as f:
161         f.write("Deliverables:\n- main_fastapi_example.py\n- run_log.txt\n- console_screenshot.png\n\nRun `python transport_demo.py` to re-run")
162
163     log(f"Saved deliverables in {out_dir}")
164
165     return {
166         "logfile": logfile,
167         "screenshot": screenshot_path,
168         "out_dir": out_dir,
169         "log_lines": log_lines
170     }
171
172 except Exception as e:
173     tb = traceback.format_exc()
174     errpath = os.path.join(out_dir, "error.txt")
175     with open(errpath, "w") as f:
176         f.write(tb)
177     log("ERROR during demo run: " + str(e))
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OUTPUT:



The image shows a VS Code editor window with a Python script named `transport_demo.py` and its terminal output. The script defines a function `run_demo_and_save_outputs` that performs several tasks: creating a backend instance, registering vehicles, adding location updates, detecting anomalies, finding nearest vehicles, and generating a trip summary. The terminal output shows the execution of the script, with timestamps and detailed logs for each step. The output also includes a list of created files and a summary of the results.

```
def run_demo_and_save_outputs(out_dir="outputs"):
    """
    :param out_dir: log_lines
    """
    log_lines = []
    try:
        # Create backend instance
        ts = datetime.datetime.utcnow().isoformat()
        [2025-11-11T09:58:46.557328] Created backend instance.
        [2025-11-11T09:58:46.558245] Registered vehicles V100, V200, V300.
        c:\Users\sathu\transport_demo.py:31: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version
        ts = datetime.datetime.utcnow().isoformat()
        [2025-11-11T09:58:46.559387] Added location updates.
        [2025-11-11T09:58:46.560685] Detected 0 anomalies for V100.
        [2025-11-11T09:58:46.560910] Detected 0 anomalies for V200.
        [2025-11-11T09:58:46.561122] Detected 0 anomalies for V300.
        [2025-11-11T09:58:46.561831] Nearest vehicles to (17.585,78.585): [('V200', 0.154), ('V300', 14.595), ('V100', 15.521)]
        [2025-11-11T09:58:46.562155] Mock AI summary for V200: Trip summary: 5 location points from 2025-11-11T09:58:46.559318 to 2025-11-11T09:58:46.562155. Recommend checking route efficiency and safety if speeds e...
        [2025-11-11T09:58:46.721519] Saved deliverables in ./outputs

    except Exception as e:
        tb = traceback.format_exc()
        errpath = os.path.join(out_dir, "error.txt")
        with open(errpath, "w") as f:
            f.write(tb)

Created files:
console_screenshot.png
main_fastapi_example.py
README.txt
run_log.txt
PS C:\Users\sathu> C:\Users\sathu\anaconda3\Scripts\activate
```

```
[2025-11-11T09:58:46.557328] Created backend instance.
[2025-11-11T09:58:46.558245] Registered vehicles V100, V200, V300.
[2025-11-11T09:58:46.559318] Added location updates.
[2025-11-11T09:58:46.560685] Detected 0 anomalies for V100.
[2025-11-11T09:58:46.560910] Detected 0 anomalies for V200.
[2025-11-11T09:58:46.561122] Detected 0 anomalies for V300.
[2025-11-11T09:58:46.561831] Nearest vehicles: [('V200', 0.154), ('V300', 14.595), ('V100', 15.521)]
[2025-11-11T09:58:46.562155] Mock AI summary generated.
[2025-11-11T09:58:46.721519] Saved deliverables in ./outputs
```

THE ABOVE IMAGE IS CONSOLE_SCREENSHOT

EXPLANATION:

The code was tested by registering three vehicles, adding location updates, running anomaly detection, generating a trip summary, and finding the nearest vehicle. The program automatically created output files:

run_log.txt – log of all test steps

console_screenshot.png – screenshot-style output image

main_fastapi_example.py – sample API backend

README.txt – description file